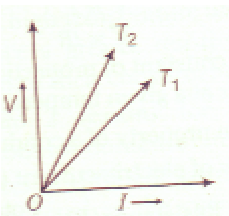
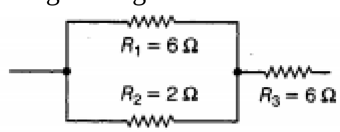
ELECTRICITY -04

Class 10 - Science

1. Resistance of an incandescent filament of a bulb is comparatively much more than that when it is at room temperature. Why? **[1]**
2. A student using the same two resistors, ammeter, voltmeter, and battery, makes two circuits connecting the two resistors first in series and second in parallel. If the ammeter and voltmeter readings in both the cases be I_1, I_2, and V_1, V_2, respectively. Write his observations. **[1]**
3. How is a voltmeter connected in the circuit to measure the potential difference between two points? **[1]**
4. Write the relation between heat energy produced in a conductor when a potential difference V is applied across its terminals and a current I flows through it for time t. **[1]**
5. What is SI unit of resistivity ? **[1]**
6. Define electric power. **[1]**
7. Which combination is used for connecting the device in the circuit to measure the potential difference across two points? **[1]**
8. The voltage-current (V-I) graph of a metallic circuit at two different temperatures T_1 and T_2 is shown in figure. Which of the two temperatures is higher and why? **[1]**



9. Why do electricians wear rubber hand gloves while working? **[1]**
10. Why closed path is required for the flow of current? **[1]**
11. Electric current flows through a metallic conductor from its one end A to other end B. Which end of the conductor is at higher potential? Why? **[1]**
12. What is the direction of electronic current ? **[1]**
13. Name the practical unit of power and state its relation with the S.I. unit. **[1]**
14. What is the resistance of an (i) ideal ammeter and (ii) ideal voltmeter? **[1]**
15. Which among the following is the correct way connecting ammeter and voltmeter in the circuit to determine the equivalent resistance of two resistors in series? **[1]**
16. The given figure shows three resistors **[1]**



Find the combined resistance.

17. Find the minimum resistance that can be made using five resistors each of $\frac{1}{5}\Omega$. **[1]**

18. How many joule are there in 1 kWh? **[1]**

19. Write the formula of electric power (P) in terms of: **[1]**

a. Potential difference (V) and current (I).

b. Current (I) and resistance (R).

c. Potential difference (V) and resistance (R).

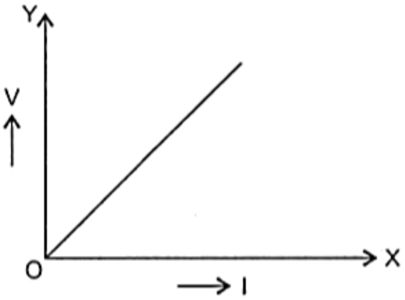
20. A wire of resistivity ρ is pulled to double its length. What will be its new resistivity? **[1]**

21. Why is an ammeter connected in series in an electric circuit? **[1]**

22. Define the term "electrical resistivity" of a material. **[1]**

23. A charge of 1500 C flows through an electric circuit in 2 min and 30 s. Calculate the magnitude of current in the circuit. **[1]**

24. The given figure shows a graph having potential difference (V) along the Y-axis and electric current (I) along the X-axis. Name the physical quantity represented by the slope of this graph. **[1]**



25. Which substance is used for making resistance coil of electric heater and why? **[1]**

26. The potential difference between the terminals of an electric heater is 60 V when it draws a current of 4 A from the source. What current will the heater draw if the potential difference is increased to 120 V? **[1]**

27. Out of the two wires P and Q shown below which one has greater resistance? Justify it. **[1]**

28. Calculate the number of electrons that must pass through the cross-section of a conductor in 1 s for 1 A current to flow through it. **[1]**

29. An electric bulb is connected to a 220 V generator. The current is 0.50 A. What is the power of the bulb? **[1]**

30. A current of 0.5 A is drawn by a filament of an electric bulb for 10 minutes. Find the amount of electric charge that flows through the circuit. **[1]**

31. A torch bulb when cold has a resistance of 2.5 Ω. It draws a current 450 mA, when connected to a 6 V battery and glows brightly. Calculate the resistance of the bulb when glowing and explain the reason for the difference in resistance. **[1]**

32. An electric refrigerator rated 400 W operates 8 hour/day. What is the cost of the energy to operate it for 30 days at Rs 3.00 per kW h? **[1]**

33. You have two metallic wires of resistances 6Ω and 3Ω. How will you connect these wires to get an effective resistance of 2Ω? **[1]**

34. A battery of 9V is connected in series with resistors of 0.2Ω, 0.3Ω, 0.4Ω, 0.5Ω and 12Ω respectively. How much current would flow through the 12Ω resistor? **[1]**

35. The resistance of a resistor is kept constant and the potential difference across its two ends is decreased to half of its former value. State the change that will occur in the current flowing through it. **[1]**

36. Define the term “volt”? **[1]**

37. What is the formula of **[1]**

a. Resistance (R) of an electric appliance,

b. Safe current (I) in terms of power rating (P) and voltage rating (V).

38. Which of the following terms does not represent electrical power in the circuit? **[1]**

(a) I^2R;

(b) IR^2;

(c) VI

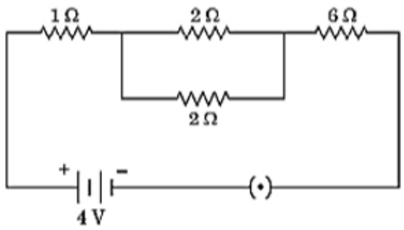
(d) $\frac{V^2}{R}$?

39. When two ends of a metallic wire are connected across the terminals of a cell, then some potential difference is set up between its ends. In which direction electrons are flowing through the conductors? **[1]**

40. Which physical quantity remains constant when resistance are connected in parallel? **[1]**

41. Define resistance of a conductor. **[1]**

42. Find the current flowing through the following electric circuit: **[1]**



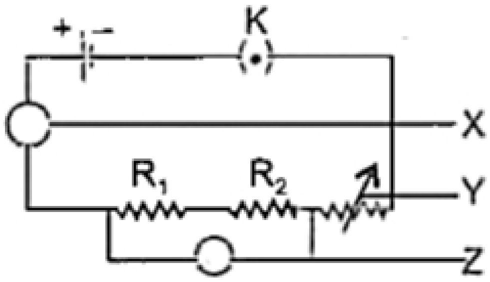
43. An electric iron of resistance 20 Ω takes a current of 5A. Calculate the heat developed in 30s. **[1]**

44. Define 1 volt potential difference. **[1]**

45. The resistance of a wire of length 150 cm and of uniform area of cross-section 0.015 cm^2, is found to be 3.0 Ω. Calculate the specific resistance of the wire. **[1]**

46. Why are coils of electric toasters and electric irons made of an alloy rather than a pure metal? **[1]**

47. Identify the X, Y and Z in the circuit given below: **[1]**



48. Write the formula for the equivalent resistance (R) when three resistors R_1, R_2 and R_3 are connected in **[1]**

a. series

b. parallel.

49. What do you understand by the term fuse in an electric circuit? **[1]**

50. State Ohm's law. **[1]**

51. Why is a series arrangement not used for connecting domestic electrical appliances in a circuit? **[1]**

52. Write the formula for the heat produced (H) when a current (I) is passed through a conductor of resistance (R) for (t) second. **[1]**

53. What happens to the other bulbs in a series circuit, if one bulb blows off? **[1]**

54. What constitutes the current ? **[1]**

55. Why is the tungsten used almost exclusively for filament of electric lamps? Explain. **[1]**

56. An electric bulb of resistance 44 Ω draws a current of 5.0 A. Calculate the line voltage. **[1]**

57. How much work is done in moving a charge of 2 C across two points having a potential difference 12 V? **[1]**

58. What is a galvanometer? **[1]**

59. Name the instrument used to detect the presence of a current in a circuit. **[1]**

60. How does the resistance of a wire vary with its area of cross-section? Explain. **[1]**

61. Write an expression for current **I**, if a charge **Q** flows through the cross-section of conductor in time **t**. **[1]**

62. What is a super conductor? Give two examples. **[1]**

63. What does an electric circuit mean? **[1]**

64. Define conductors and insulators? **[1]**

65. Draw a circuit diagram using a battery of two cells, two resistors of 3 Ω each connected in series, a plug key and a rheostat. **[1]**

66. How much work is done in moving a charge of 1.5 C across two points having a potential difference of 4 V? **[1]**

67. Let the resistance of an electrical device remain constant, while the potential difference across its two ends decreases to one fourth of its initial value. What change will occur in the current through it? State the law which helps us in solving the above stated question. **[1]**

68. What is measured by an ammeter ? **[1]**

69. Define resistivity. **[1]**

70. The following table gives the value of electrical resistivity of some materials: **[1]**

Material	**Copper**	**Silver**	**Constant**
Electrical resistivity (in Ω-m)	1.62×10^{-8}	1.6×10^{-8}	49×10^{-8}

Which one is the best conductor of electricity out of them?

71. What is SI unit of resistance ? **[1]**

72. In how much time will a bulb of 100 W consume the energy of 2 kWh? **[1]**

73. A uniform wire with a resistance of 32 Ω is divided into four equal parts and they are joined in parallel. Calculate the equivalent resistance of the parallel combination. **[1]**

74. Three resistors of 6Ω, 4Ω and 4Ω are connected together so that the total resistance is 8Ω. Draw a diagram to show this arrangement and give reason to justify your answer. **[1]**

75. Define the unit of current. **[1]**